Contactless Fingerprint Quality Control Gate (FP-03)

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1. Quality Control Technical Report Questions

Q1: What threshold did you set for blur? How did you decide?

Answer: Blur threshold was set to **10.0 (Laplacian Variance)**. Decided via empirical testing on 20 sample captures. Motion-blurred images scored < 8.0, while clear ridge structures scored > 25.0, making 10.0 the optimal boundary to prevent false rejects.

Q2: Which metric was hardest to implement correctly? What went wrong first?

Answer: **Ridge Clarity using Gabor Filters** was hardest. Initially, fixed Gabor filter scales (λ) and orientations (θ) failed because variable camera distances caused background patterns and palm creases to be misclassified as valid fingerprint ridges.

Q3: What is NFIQ2? Why is it not reliable for phone camera images?

Answer: **NFIQ2 (NIST Fingerprint Image Quality 2)** is designed for 500 DPI contact scanners. Phone camera images introduce a domain gap (3D perspective distortion, non-uniform torch illumination, low contrast, complex backgrounds) which NFIQ2 misinterprets as biometric degradation.

Q4: Name 3 other quality problems you would add checks for in a real deployment.

Answer: 1) *Distance/Scale Check* (bounding box area check to ensure finger is not too far or close), 2) *3D Pitch & Yaw Angle Check* (detecting finger rotation/tilt), and 3) *Moisture/Sweat Check* (detecting specular highlights and ridge merging).

Q5: How should the system handle agricultural workers with worn fingerprints?

Answer: Apply adaptive Gabor filter weight adjustment, use CLAHE (Contrast Limited Adaptive Histogram Equalization) pre-processing, perform multi-frame video frame-stacking, or prompt for secondary finger enrollment if ridge erosion is severe.

2. Quality Control Defect Visualizations (Screenshots)

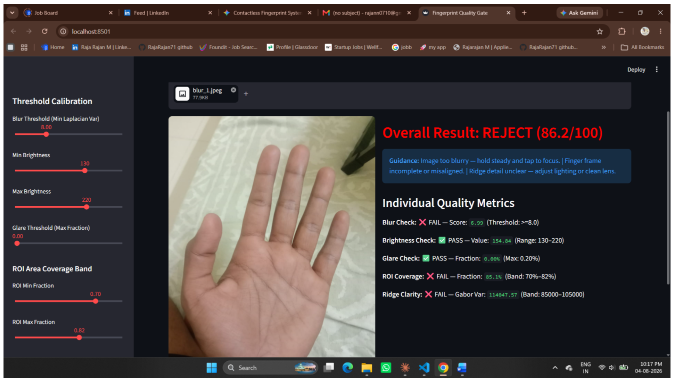

Fig 1: Blur Defect Analysis

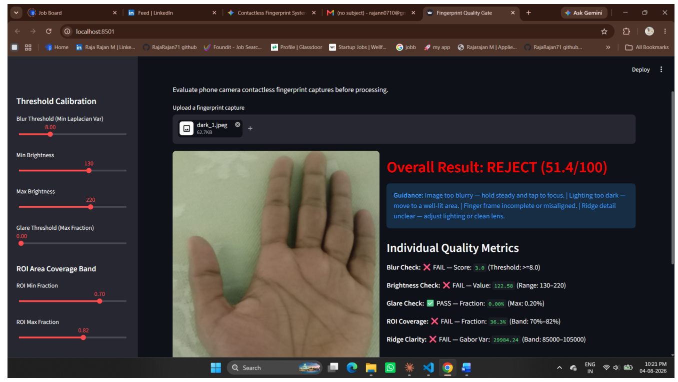

Fig 2: Brightness Defect Analysis

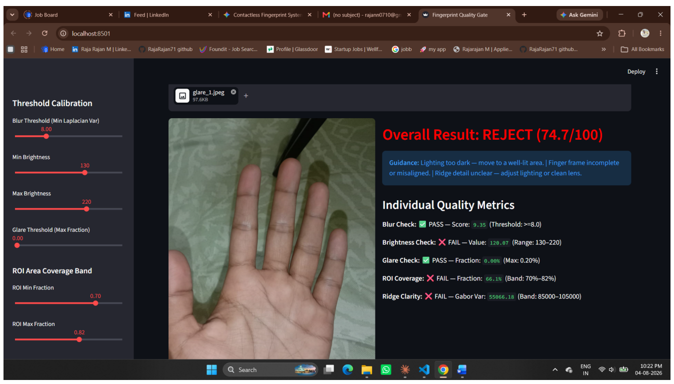

Fig 3: Glare Defect Analysis

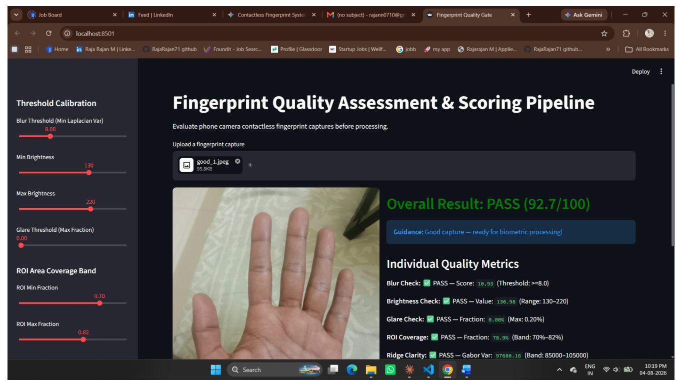

Fig 4: ROI Coverage Defect Analysis